Supplementary Material

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I. Supplementary Theoretical Analysis

A. Expressive Power of Subgraph Neural Network

Following prior work [1, 2], the Weisfeiler-Lehman test [3] is used as the evaluation criterion. According to [4], for graphs with unweighted edges, 1-WL and 2-WL are known to have the same discriminative power. The GIN model [1] has demonstrated that message passing GNNs are bounded in their expressive power by the 1-WL test:

Lemma I.1 ([1]). Let $f : \mathcal{G} \to \mathbb{R}^d$ be a message passing GNN. With a sufficient number of GNN layers, f maps any graphs $\mathcal{G}_1$ and $\mathcal{G}_2$ that the Weisfeiler-Lehman test of isomorphism decides as non-isomorphic, to different embeddings if the update function of each layer and the readout function are injective.

Compared to message passing GNNs [1, 5–7], subgraph neural network exhibits greater expressive power. Subgraph neural network computes subgraph representations by processing the extracted ego networks, learning intermediate representations for each node within the subgraph. A pooling layer is then applied to aggregate these node representations into a final subgraph representation, which ultimately serves as the representation for the central node of the ego network. In subgraph neural networks, the same node will have different representations depending on the ego network in which it appears. In contrast, in message passing GNNs, a node's representation remains uniform, regardless of the various subgraphs it contributes to. In [8], the subgraph neural network is proved to be more powerful than 2-WL test:

Lemma I.2 ([8]). Consider all pairs of n-sized r-regular graphs, where $3 \leq r < (2logn)^{\frac{1}{2}}$.For any small constant $\epsilon > 0$, if a subgraph neural network fulfills:

- the message passing and readout function are injective;
- the input is from a countable space,

it can distinguish almost all $(1 - o(1))$ such pairs of graphs using at most $\lceil(\frac{1}{2} + \epsilon)\frac{\log n}{\log(r-1-\epsilon)}\rceil$-hop ego networks and $\lceil\epsilon\frac{\log n}{\log(r-1-\epsilon)}\rceil$-layer message passing.

Since the subgraph neural network can distinguish nearly all r-regular graphs that the 1-WL test fails to distinguish [2, 8], it is strictly more powerful than message passing GNNs. Furthermore, this implies that subgraph neural network does not require ego networks with excessively large hop numbers (approximately $\frac{1}{2}\frac{\log n}{\log(r-1-\epsilon)}$) to achieve greater expressive power.

B. Prompt Capability

Lemma I.3 (Prompt effectiveness guarantee [9]). For a series of graphs $\mathcal{D} = \{(\mathcal{G}_1, y_1), \ldots, (\mathcal{G}_n, y_n)\}$, a pre-trained GNN model $f_{W_0} = (f_1, \cdots, f_n)$ with each $f_i(X_i) = \mathbb{1} \cdot S_i X_i W_0$, under the non-degeneracy condition [10], a linear projection head θ, and a learnable vector $\mathbf{v}$, there exists $\mathcal{Y}' = \{y'_1, \ldots, y'_n\}$ for which we have

$$\begin{aligned} l_p &= \min_{\mathbf{v},\theta} \sum_{i=1}^{n} (f_i(X_i + \mathbb{1}^T \cdot \mathbf{v}) \cdot \theta - y_i)^2 < l_{ft} \\ &= \min_{\tilde{f}\in\tilde{\mathcal{F}},\theta} \sum_{i=1}^{n} (\tilde{f}_i(X_i) \cdot \theta - y_i)^2 \end{aligned} \tag{1}$$

when $y_1 = y'_1, \ldots, y_n = y'_n$. Here $\mathbb{1}$ is a vector with all items equals to 1 and with a proper size that changes from line to line, and $\tilde{\mathcal{F}}$ is defined as

$$\begin{aligned} \tilde{\mathcal{F}} = \Big\{ \tilde{f} = (\tilde{f}_1, \cdots, \tilde{f}_n) : \tilde{f}_i = \mathbb{1} \cdot S_i X_i W, 1 \leq i \leq n \\ \text{with } \forall\, (W - W_0) \Big\}. \end{aligned} \tag{2}$$

Lemma I.3 [9] provides the effectiveness guarantee for learnable prompt vector, which is applied in downstream centrality prediction tasks.

C. Detailed Proof of Theorem 1

Theorem 1. *For a series of graphs $\mathcal{D} = \{(\mathcal{G}_1, y_1), \ldots, (\mathcal{G}_n, y_n)\}$, a pre-trained GNN model $f_{W_0} = (f_1, \cdots, f_n)$ with each $f_i(X_i) = \mathbb{1} \cdot S_i X_i W_0$, and an arbitrary pattern graph $\mathcal{G}_p$, under the non-degeneracy condition, a linear projection head θ and a GNN $f_p \in \mathcal{F}$ for prompting, there exists $\mathcal{Y}' = \{y'_1, \ldots, y'_n\}$ for which we have*

$$\begin{aligned} l_p &= \min_{f_p\in\mathcal{F},\theta} \sum_{i=1}^{n} (f_i(X_i + \mathbb{1}^T \cdot f_p(X_p)) \cdot \theta - y_i)^2 \\ &< l_{ft} = \min_{\tilde{f}\in\tilde{\mathcal{F}},\theta} \sum_{i=1}^{n} (\tilde{f}_i(X_i) \cdot \theta - y_i)^2 \end{aligned} \tag{3}$$

when $y_1 = y'_1, \ldots, y_n = y'_n$. Here $\mathbb{1}$ is a vector with all items equals to 1 and with a proper size that changes from line to line, $\tilde{\mathcal{F}}$ is defined in (6), and $\mathcal{F}$ is defined as

$$\mathcal{F} = \left\{ f_p : f_p = \mathbb{1} \cdot S_p X_p W_p, \text{with } \forall\, (W_p - X_p^T X_p) \right\}.$$

Before the proof, we first give the definition of degeneracy condition:

Definition I.1 (Degeneracy condition [10]). Given a matrix $\Phi \in \mathbb{R}^n$ where $\Phi = \{\phi_1, \ldots, \phi_n\}$, if two or more of its basis vectors ϕ_j are colinear, matrix Φ can be considered degenerate.

Based on the degeneracy condition and graph $G = (A, X, Y)$, we can derive the non-degeneracy condition for G:

1) For adjacency matrix A, it should be irreducible, which means that the given graph is a connected graph [11].
2) For feature matrix X, $X^T X$ should be a full-rank matrix.
3) For label vector Y, each $y_i \in Y$ should be distinct.

Proof. Here S_i, $1 \leq i \leq n$, and S_p is the diffusion matrix [12] of the graph $(\mathcal{G}_i, y_i)$, $1 \leq i \leq n$ and the graph $\mathcal{G}_p$ respectively, i.e.,

$$S_i = \sum_{k=0}^{\infty} \theta_{i,k} T_i^k, \ 1 \leq i \leq n, \ \text{ and } \ S_p = \sum_{k=0}^{\infty} \theta_{p,k} T_p^k, \quad (4)$$

for some properly chosen weighting coefficients $\{\theta_{i,k}\}_{1 \leq i \leq n, k \geq 1}$ and $\{\theta_{p,k}\}_{k \geq 1}$, and the properly chosen generalized transition matrix $\{T_i\}_{1 \leq i \leq n}$ and T_p that are chosen based on the corresponding graph's structure.

Now, recall Lemma I.3 [9] states that

***Lemma* A.3** (Prompt effectiveness guarantee [9])**:** For a series of graphs $\mathcal{D} = \{(\mathcal{G}_1, y_1), \ldots, (\mathcal{G}_n, y_n)\}$, a pre-trained GNN model $f_{W_0} = (f_1, \cdots, f_n)$ with each $f_i(X_i) = \mathbb{1} \cdot S_i X_i W_0$, under the non-degeneracy condition [10], a linear projection head θ, and a learnable vector $\mathbf{v}$, there exists $\mathcal{Y}' = \{y_1', \ldots, y_n'\}$ for which we have

$$\begin{aligned} l_p &= \min_{\mathbf{v}, \theta} \sum_{i=1}^{n} (f_i(X_i + \mathbb{1}^T \cdot \mathbf{v}) \cdot \theta - y_i)^2 < l_{ft} \\ &= \min_{\tilde{f} \in \tilde{\mathcal{F}}, \theta} \sum_{i=1}^{n} (\tilde{f}_i(X_i) \cdot \theta - y_i)^2 \end{aligned} \quad (5)$$

when $y_1 = y_1', \ldots, y_n = y_n'$. Here $\mathbb{1}$ is a vector with all items equals to 1 and with a proper size that changes from line to line, and $\tilde{\mathcal{F}}$ is defined as

$$\begin{aligned} \tilde{\mathcal{F}} = \Big\{ \tilde{f} = (\tilde{f}_1, \cdots, \tilde{f}_n) : \tilde{f}_i = \mathbb{1} \cdot S_i X_i W, 1 \leq i \leq n \\ \text{with } \forall \ (W - W_0) \Big\}. \end{aligned} \quad (6)$$

Therefore, to conclude Theorem 1, we only need to show that for arbitrary given $\mathcal{Y} = \{y_1, \ldots, y_n\}$, we have

$$\begin{aligned} &\min_{f_p \in \mathcal{F}, \theta} \sum_{i=1}^{n} (f_i(X_i + \mathbb{1}^T \cdot f_p(X_p)) \cdot \theta - y_i)^2 \\ &= \min_{\mathbf{v}, \theta} \sum_{i=1}^{n} (f_i(X_i + \mathbb{1}^T \cdot \mathbf{v}) \cdot \theta - y_i)^2, \end{aligned} \quad (7)$$

or equivalently, if we assume each X_i has p_0 columns, then, for arbitrary $\mathbf{v} \in \mathbb{R}^{1 \times p_0}$, there exists a matrix $W_p \in \mathbb{R}^{p_0 \times p_0}$ such that

$$\mathbf{v} = \mathbb{1} \cdot S_p X_p W_p \triangleq b_p \cdot W_p. \quad (8)$$

To proof (8), notice that $b_p \in \mathbb{R}^{1 \times p_0}$ and according to Gram-Schmidt orthogonalization process, there exists a base $\{b_1, \cdots, b_{p_0}\}$ such that $b_i = (b_{i1}, \cdots, b_{ip_0}) \in \mathbb{R}^{1 \times p_0}$ for $1 \leq i \leq n$, $b_1 = b_p / \|b_p\|_2$ and

$$\tilde{W} = (b_1^T, \cdots, b_{p_0}^T) \quad (9)$$

is a orthonormal matrix. Now, denote $e_1 = (1, 0, \cdots, 0) \in \mathbb{R}^{1 \times p_0}$. Apparently,

$$W_p = \frac{1}{\|b_p\|_2} \cdot \left[b_1^T \cdot (\mathbf{v} - e_1) + \tilde{W} \right] \quad (10)$$

is one of the solutions of equation (8). Hence, for arbitrary $\mathbf{v} \in \mathbb{R}^{1 \times p_0}$, equation (8) has a solution $W_p \in \mathbb{R}^{p_0 \times p_0}$, and by applying lemma I.3, further leads to Theorem 1.

□

II. Detailed Proof of Theorem 2

Theorem 2. *Let $G = (V, E)$ be a connected graph with normalized Laplacian $\mathbf{L}$ and second smallest eigenvalue $\lambda_2 > 0$. Let $\{G_i = (V_i, E_i)\}_{i \in V}$ be the set of k-hop ego-network subgraphs for an SNN, each with normalized Laplacian $\mathbf{L}_i$ and second smallest eigenvalue $\lambda_{2,i}$. Let $\lambda_{2,min} = \min_{i \in V} \{\lambda_{2,i}\}$. The convergence suppression rate to an over-smoothed state $\lambda_{2,min} \geq \lambda_2$.*

We quantify Over-smoothing using the Dirichlet energy [13] of the node embeddings $\mathbf{H} \in \mathbb{R}^{N \times d}$ on a graph G:

$$E(\mathbf{H}) = \frac{1}{2} \operatorname{Tr}(\mathbf{H}^T \mathbf{L} \mathbf{H}) \quad (11)$$

where $\mathbf{L} = \mathbf{I} - \tilde{\mathbf{A}}$ is the normalized Laplacian of the graph, and $\tilde{\mathbf{A}} = \mathbf{D}^{-1/2}(\mathbf{A} + \mathbf{I})\mathbf{D}^{-1/2}$ is the normalized adjacency matrix with self-loops. A lower Dirichlet energy signifies that the node embeddings are more similar across connected nodes.

Proof. For simplicity, we consider a linear GNN without non-linearities, as the core of over-smoothing lies in the propagation operator $\tilde{\mathbf{A}}$. In a standard GNN, the propagation rule is:

$$\mathbf{H}^{(l+1)} = \tilde{\mathbf{A}} \mathbf{H}^{(l)} \quad (12)$$

After L layers, the node embeddings are $\mathbf{H}^{(L)} = \tilde{\mathbf{A}}^L \mathbf{H}^{(0)}$. The Dirichlet energy at layer L is:

$$\begin{aligned} E(\mathbf{H}^{(L)}) &= \frac{1}{2} \operatorname{Tr}((\mathbf{H}^{(L)})^T \mathbf{L} \mathbf{H}^{(L)}) \\ &= \frac{1}{2} \operatorname{Tr}((\mathbf{H}^{(0)})^T (\tilde{\mathbf{A}}^L)^T \mathbf{L} \tilde{\mathbf{A}}^L \mathbf{H}^{(0)}) \end{aligned} \quad (13)$$

Since $\mathbf{L} = \mathbf{I} - \tilde{\mathbf{A}}$, and $\mathbf{L}$ and $\tilde{\mathbf{A}}$ are symmetric and share the same eigenvectors, we can analyze their eigenvalues. Let the eigenvalues of $\mathbf{L}$ be $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_N \leq 2$. The eigenvalues of $\tilde{\mathbf{A}}$ are $1 = \mu_1 \geq \mu_2 \geq \cdots \geq \mu_N \geq -1$, where $\mu_j = 1 - \lambda_j$. The rate of convergence to the smoothed state is dictated by $\mu_2 = 1 - \lambda_2$. A smaller λ_2 (a smaller spectral gap) leads to a μ_2 closer to 1, causing faster convergence.

In an SNN, message passing occurs independently within each ego-network G_i. Let $\mathbf{H}_i^{(0)}$ be the initial features of nodes

in subgraph G_i. The propagation is governed by its local normalized adjacency matrix, $\tilde{\mathbf{A}}_i$:

$$\mathbf{H}_i^{(l+1)} = \tilde{\mathbf{A}}_i \mathbf{H}_i^{(l)} \tag{14}$$

After L layers, the embeddings are $\mathbf{H}_i^{(L)} = \tilde{\mathbf{A}}_i^L \mathbf{H}_i^{(0)}$. The final representation for the central node v_i is computed by a readout function over these local embeddings:

$$\mathbf{h}_i^{(L)} = \text{READOUT}(\mathbf{H}_i^{(L)}) \tag{15}$$

The over-smoothing analysis must now be applied locally to each subgraph G_i. The Dirichlet energy of the embeddings within subgraph G_i is:

$$E_i(\mathbf{H}_i^{(L)}) = \frac{1}{2}\,\text{Tr}((\mathbf{H}_i^{(L)})^T \mathbf{L}_i \mathbf{H}_i^{(L)}) \tag{16}$$

The rate of convergence to an over-smoothed state for the nodes within this subgraph is determined by the spectral gap of G_i, characterized by its second smallest Laplacian eigenvalue, $\lambda_{2,i}$.

Lemma II.1 (Cheeger Inequality [14]). For a connected graph G, with its Cheeger isoperimetric constant k_G, we have

$$2k_G \geq \lambda_{2,G} \geq \frac{k_G^2}{2} \tag{17}$$

We argue that ego-networks are structurally better connected than the global graph for the following reason:

The global Cheeger constant, k_G, is defined by the single most significant bottleneck found anywhere across the entire graph [14].

An ego-network, G_i, is a small, localized neighborhood around a single node.

It is improbable that a subgraph like G_i will happen to contain the entire graph's primary bottleneck. Because the bottleneck is a global property, the best possible sparse cut that can be found within the limited confines of G_i will almost always be denser than the global one. This leads to the natural structural condition that for a typical subgraph, its Cheeger constant is greater than or equal to the global one: $k_{G_i} \geq k_G$.

Applying this condition through the Cheeger inequality, we establish our desired relationship:

$$\lambda_{2,G_i} \geq \frac{k_{G_i}^2}{2} \geq \frac{k_G^2}{2} \tag{18}$$

The inequality also gives $2k_G \geq \lambda_{2,G}$, which implies $k_G^2 \geq \lambda_{2,G}^2/4$. Therefore, for any ego-network G_i, there exists a constant c that $\lambda_{2,G_i} \geq k_{G_i}^2/2 \geq c^2 k_G^2/2 \geq 2k_G \geq \lambda_{2,G}$, where $k_{G_i} \geq c \cdot k_G \geq \frac{4}{c}$.

This establishes our claim that $\lambda_{2,\text{min}} \geq \lambda_2$, which means the term $(1 - \lambda_{2,\text{min}})^L$ approaches zero more slowly than $(1 - \lambda_2)^L$. By confining message passing to these better-conditioned subgraphs, the SNN architecture significantly reduces the rate of convergence to an over-smoothed state. □

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